Agile Presentation

CS-250 Software Development Lifecycle

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Agile Roles

Scrum Master

- They act as a servant leader and are responsible for making the Scrum team as effective as possible, and do not boss others around.
- They plan and lead the Scrum meetings such as the Daily Scrum Meetings, Sprint Planning, and Retrospectives.
- They protect the Scrum team by removing and working out any issues/impediments that are in the way of the team.

(Coursera Staff, 2025a)

Agile Roles

Product Owner

- They ensure to align the Scrum team with the project goals and bridge between the business needs and technical team.
- They are responsible for managing the Product Backlog to ensure work is prioritized so the most important features are done first.
- They create User Stories for the Development team to see what items are needed for the project.
- They communicate with stakeholders and customers to ensure they keep with trends and expectations for the product as well as the wants and needs of the customer.

(Coursera Staff, 2025a)

Agile Roles

Developers

- They are the people who do the work/code for the product and complete the tasks during the Sprint.
- They are responsible for providing a useable high-quality product by the end of the project deadline.
- They are self organized and collaborate to map out goals and find the best technical way to get them done and achieve them.

(Coursera Staff, 2025a)

Agile Roles

Testers

- They create test cases to state what the function of an item is and what its expected results will be.
- They are responsible for preventing a faulty product and keeping the user experience from becoming unpleasant.
- They try to "break" the product to find any glitches/bugs to fix issues before releasing the product out to the customers.
- They ensure that when adding new features, it does not hurt the existing security of the application.

(Coursera Staff, 2025b)

Agile Phases

A team using Agile will follow a structured repeating iteration cycle.

Planning

- The team collaborates with stakeholders to define the projects vision and prioritize the initial features of what the product will be and what it will do for the customers.
- Research is done to see what competitors are doing to ensure the best quality, and the team estimates costs of the project.

(Nguyen, 2024)

(Hartshorne, 2025)

Agile Phases

Development/Execution

- The team starts to build the product and create functional software in short cycles called Sprints.
- Daily meetings are held to address any issues/impediments that have come up, and the team provides updates for continuous momentum.

Testing

- All testing is integrated into each Sprint cycle so the team can check for bugs and ensures that the software meets functional requirements for the goals of the project.

(Nguyen, 2024)

(Hartshorne, 2025)

Agile Phases

Review

- At the end of each Sprint cycle, the team will show the working software to stakeholders who will then provide helpful feedback.
- This step helps to ensure that the software meets with requirements and is now ready to move onto more development.

Release

- In this stage, the product is deployed to the customers, and may also include beta testing as well.
- The team will examine the software to verify the code is clean and promptly addresses any bugs.

(Nguyen, 2024)

(Hartshorne, 2025)

Agile Phases

Maintenance

- After the release of the product, the development team will continue to maintain the software ensuring smooth operation.
- The team will also continue to address any bugs that must be taken care of, as well as providing any help to the customers to help with their proficiency with the product.

(Nguyen, 2024)

(Hartshorne, 2025)

Using Waterfall Model VS. Agile

Agile is best suited for projects that are constantly changing.

The SNHU Travel project used an Agile approach which let the team:

- Receive stakeholder/customer feedback throughout development to stay within the trends of travel destinations.
- Allowed the team to make changes to the application in the middle of the project without disrupting the flow of the project. For instance, changing destinations from overall travel to destinations that are detox and wellness themed.
- Code was tested regularly to ensure features work before moving onto another task.

Using Waterfall Model VS. Agile

The Waterfall approach is best suited for projects that will not have constant changes.

If the SNHU Travel project used a Waterfall Model instead:

- All features are planned at the beginning and changes are not allowed till the very end of the project. For instance, the stakeholders/customers wanted to change from overall travel to detox and wellness themed travel, but the change cannot take place till late in the project causing a delay in the release.
- Code is not tested till the end of the project, possibly leading to major bugs that now need to be fixed in a short time frame and causing the release date to be pushed back and the company to spend more money on the delayed application.

When should you use a Waterfall or Agile Approach?

Waterfall

- This process is for projects that need well-structured planning and are not expected to experience changes.
- Is best for when lots of detailed planning documentations are needed to be created.
- Follows a step-by-step process which requires one stage to be finished before starting the next one.
- Stakeholders/customers are involved at the beginning of the project to determine what is needed, but are not very involved once the project starts.

(Adobe Communications Team, 2025)

When should you use a Waterfall or Agile Approach?

Waterfall Continued

- Making changes is hard to do once a stage is done and could cause problems leading to several issues and huge money costs to fix these issues.
- The time frame and delivery of the project is determined at the beginning and you are to have one large release at the end of the project.
- Best for when you want teams that work in individual groups and receive instructions from top down.

(Adobe Communications Team, 2025)

When should you use a Waterfall or Agile Approach?

Agile

- This process is for projects that are constantly changing based on trends and customer needs/wants.
- Has planning but not a whole lot of documentations as the focus is on delivering working parts of the project.
- Follows a cycle-by-cycle process which allows the project to be broken down and improve incrementally.
- Stakeholders/customers are involved throughout the project providing constant feedback which is best for projects that are always looking at trends.

(Adobe Communications Team, 2025)

When should you use a Waterfall or Agile Approach?

Agile Continued

- Making changes is easy to do and this process welcomes new ideas which can then be implemented throughout the project.
- Problems are resolved in smaller steps, allowing the project to adapt quickly to these issues.
- The time frame and delivery of the project is estimated at the beginning, but the timeline can be adjusted at any point as you get many working parts rather than one big release at the very end.
- Best for when you want teams to work together closely and have people with diverse skill sets that communicate daily with one another.

(Adobe Communications Team, 2025)

When should you use a Waterfall or Agile Approach?

SNHU Travel Project and why Agile was the best choice.

- Throughout this project, changes were made to fit with customer needs that were changing based on trends going on with travel.
- The team communicated daily through meetings and shared information on how the tasks were moving along.
- Throughout the project, the team was able to deliver working parts of the application to showcase to stakeholders/customers and the team.

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